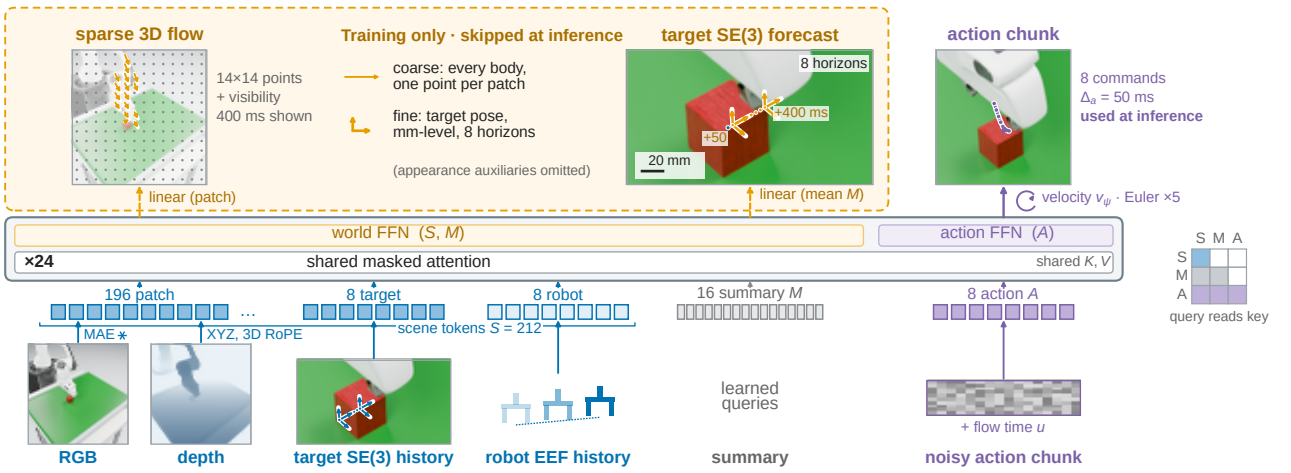

sparse 3D flow
14×14 points
+ visibility
400 ms shown
Training only · skipped at inference
coarse: every body,
one point per patch
fine: target pose,
mm-level, 8 horizons
(appearance auxiliaries omitted)
target SE(3) forecast
8 horizons
+400 ms
+50
20 mm
action chunk
8 commands
Δ_a = 50 ms
used at inference
linear (patch)
linear (mean M)
velocity v_ψ · Euler ×5
world FFN (S, M)
action FFN (A)
×24
shared masked attention
shared K, V
S M A
S
M
A
query reads key
196 patch
…
8 target
8 robot
16 summary M
8 action A
MAE ∗
XYZ, 3D RoPE
scene tokens S = 212
learned
queries
+ flow time u
RGB
depth
target SE(3) history
robot EEF history
summary
noisy action chunk